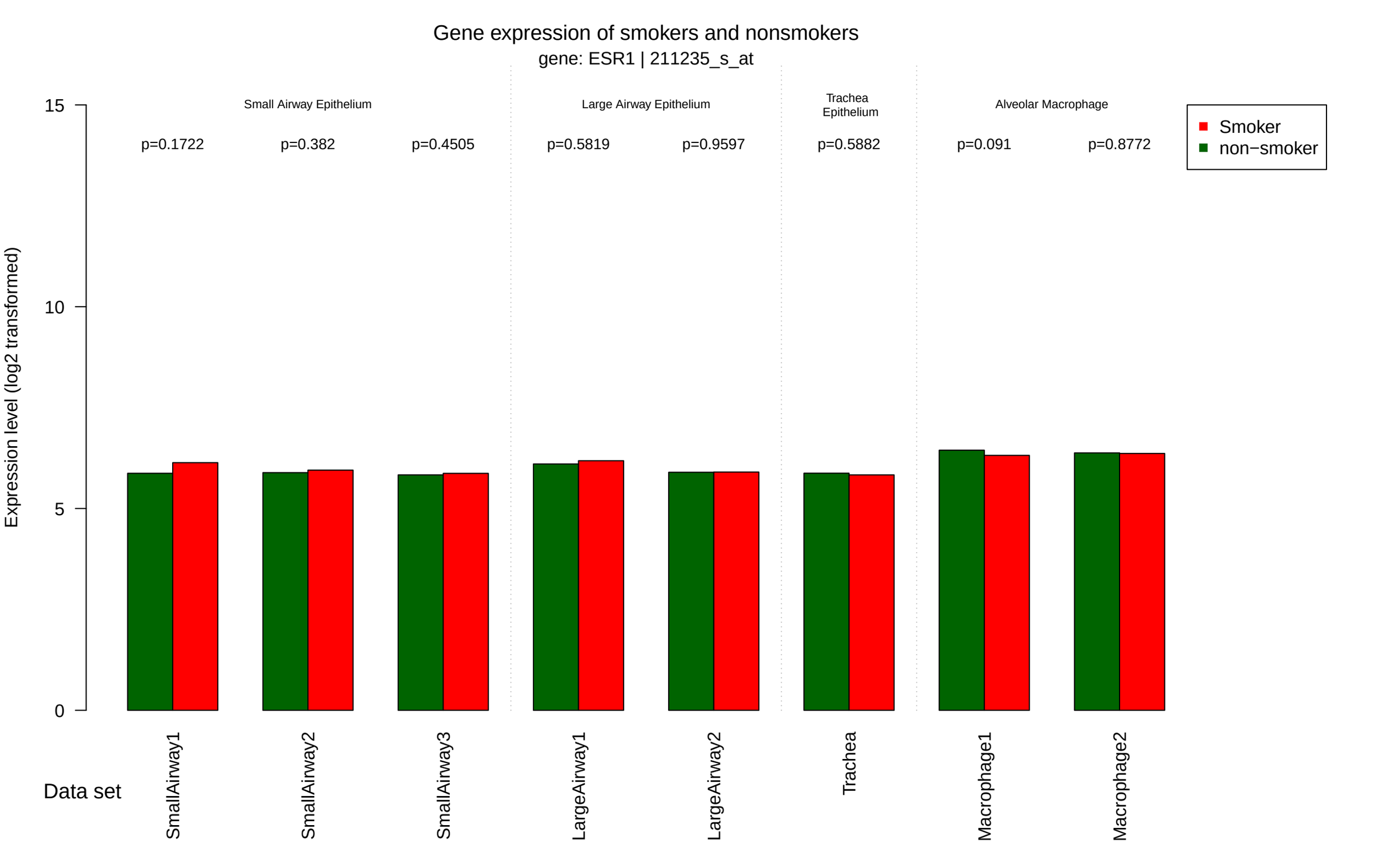
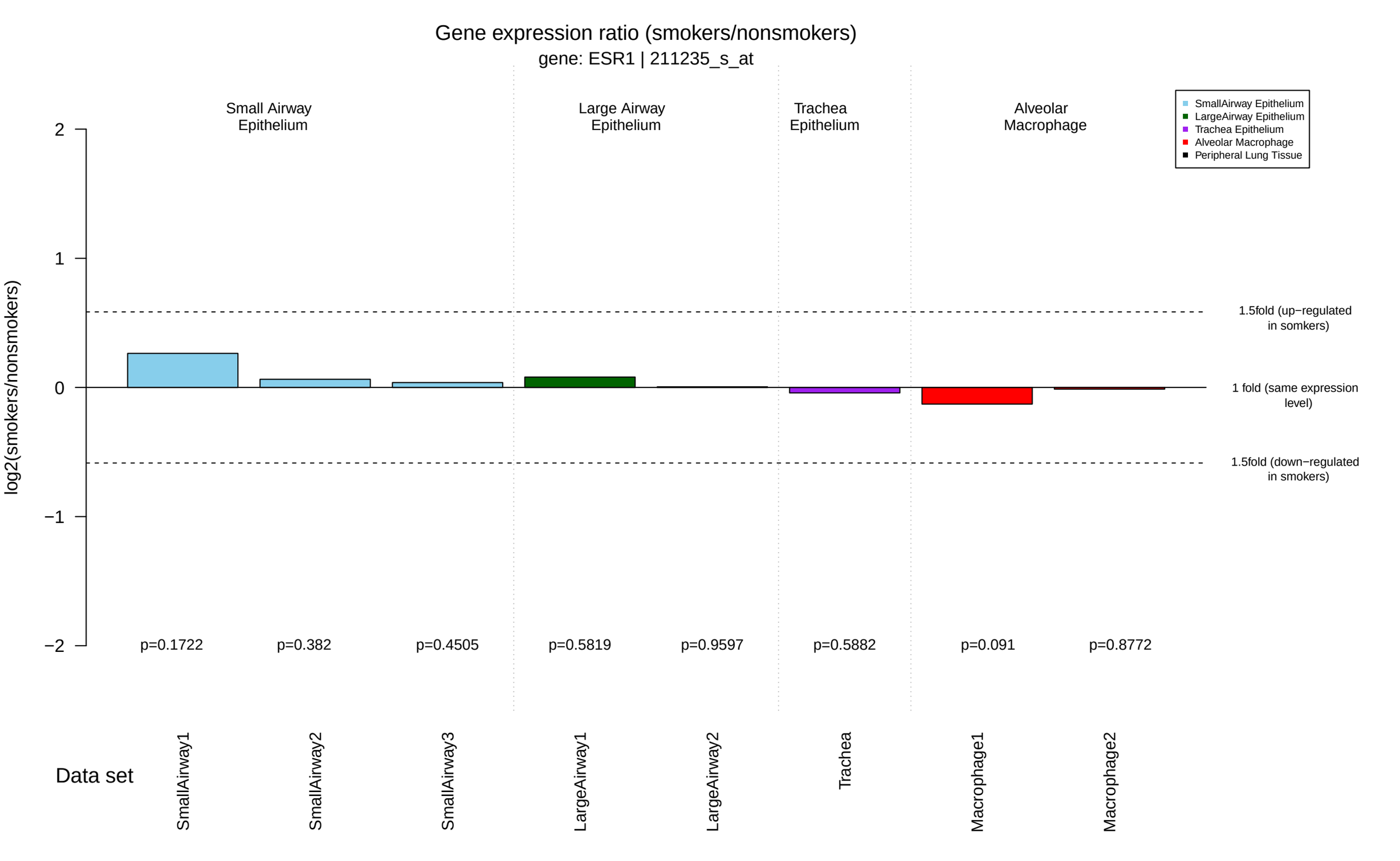
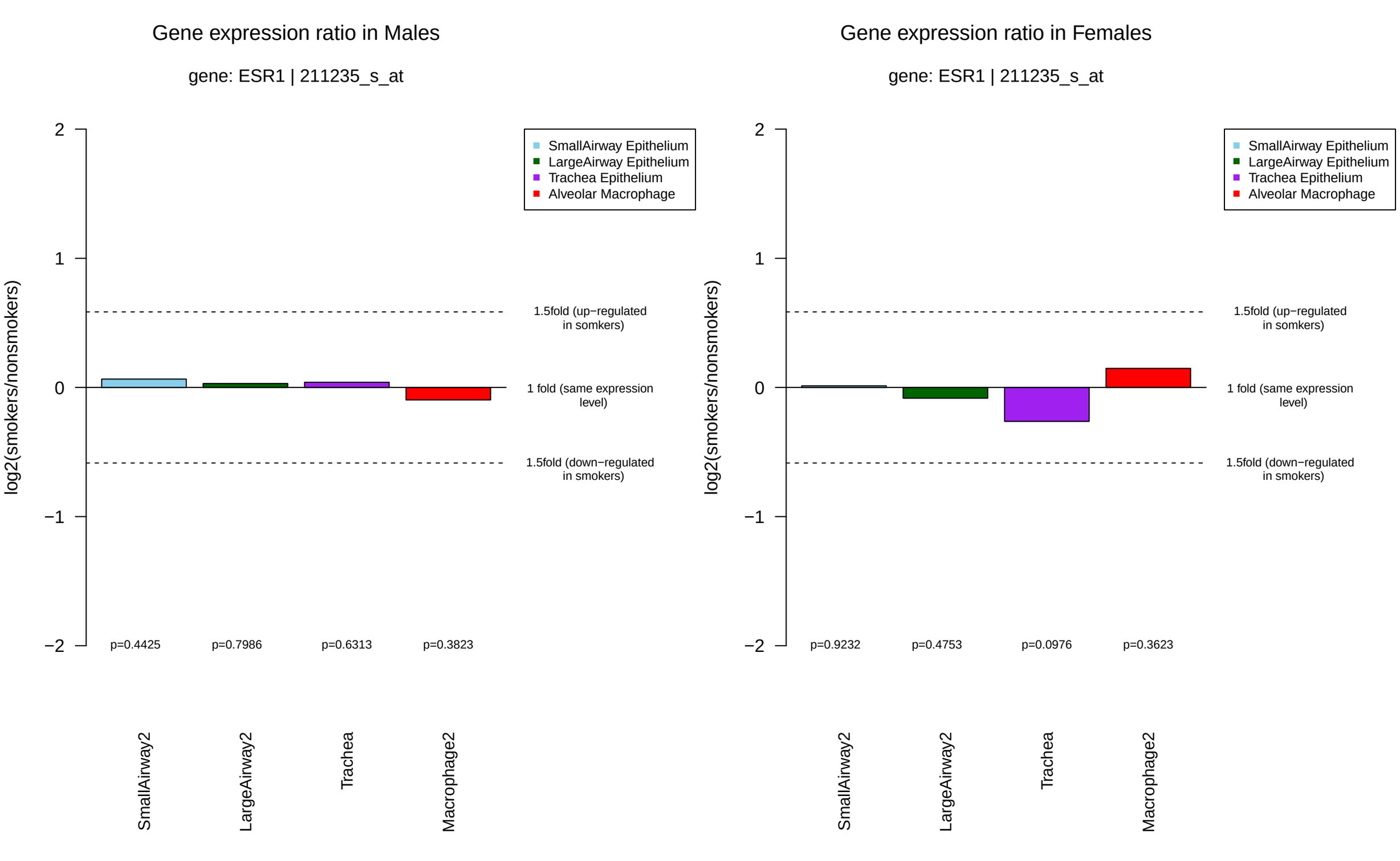
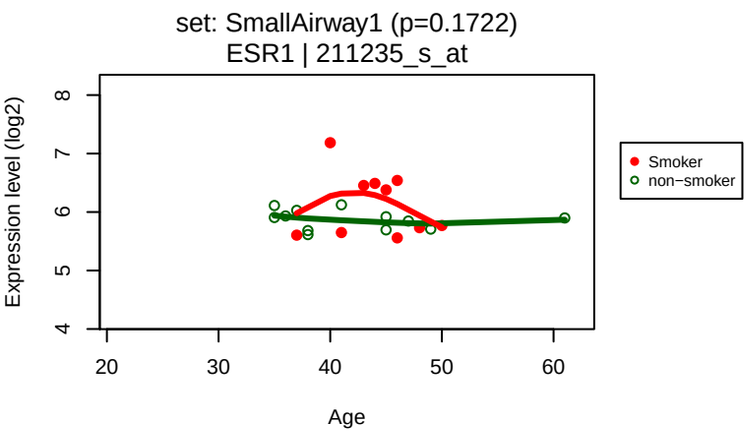
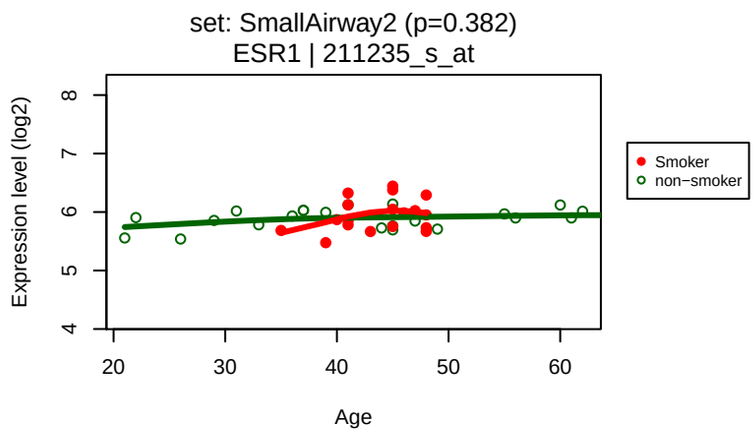
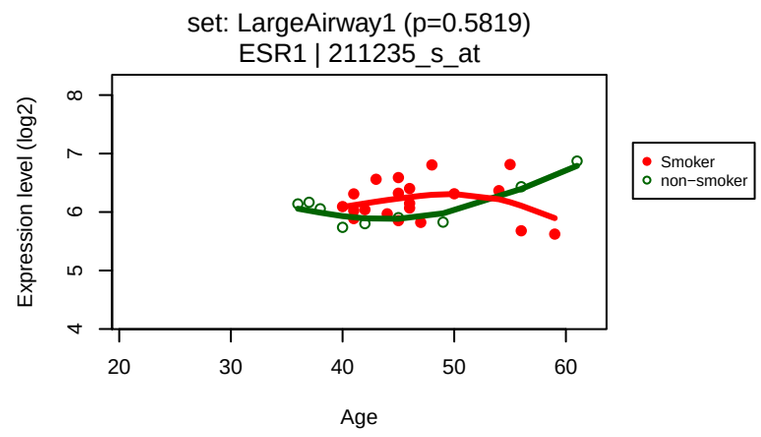

Gene expression of smokers and nonsmokers
gene: ESR1 | 211235_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1722
p=0.382
p=0.4505
p=0.5819
p=0.9597
p=0.5882
p=0.091
p=0.8772
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

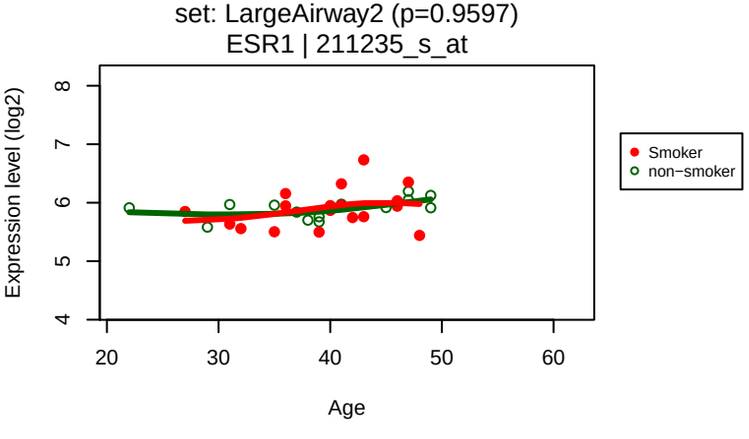

Gene expression ratio (smokers/nonsmokers)
gene: ESR1 | 211235_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1722
p=0.382
p=0.4505
p=0.5819
p=0.9597
p=0.5882
p=0.091
p=0.8772
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

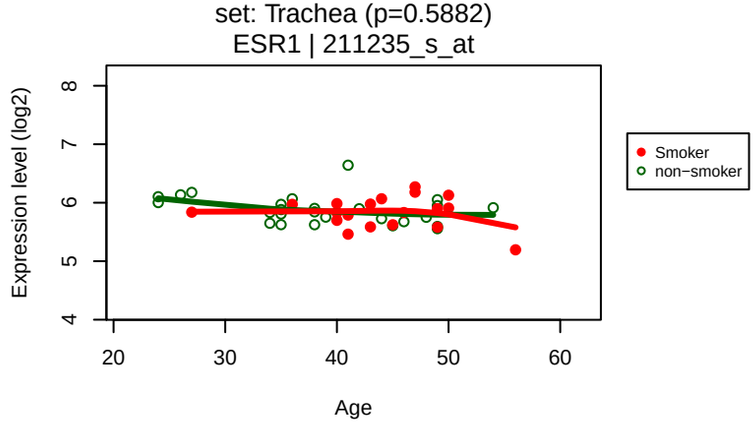

Gene expression ratio in Males
gene: ESR1 | 211235_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4425
p=0.7986
p=0.6313
p=0.3823
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ESR1 | 211235_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9232
p=0.4753
p=0.0976
p=0.3623
SmallAirway2
LargeAirway2
Trachea
Macrophage2

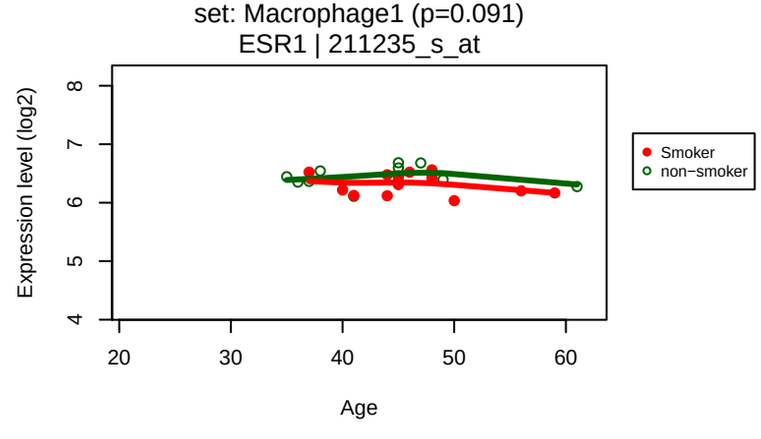

set: SmallAirway1 (p=0.1722)
ESR1 | 211235_s_at
Expression level (log2)
Age
Smoker
non−smoker

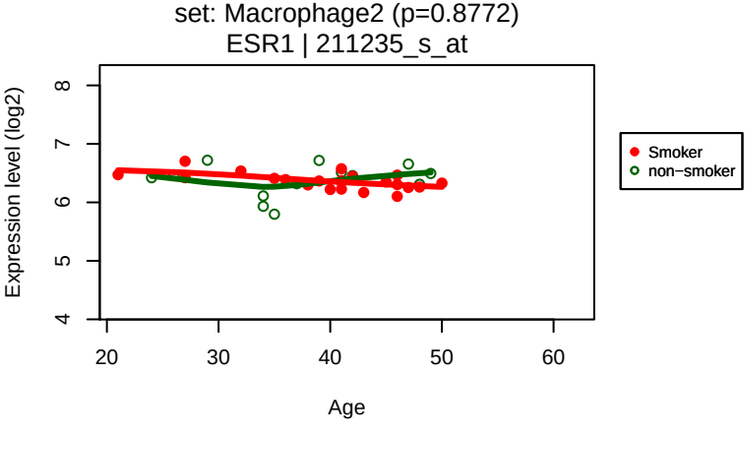

set: SmallAirway2 (p=0.382)
ESR1 | 211235_s_at
Expression level (log2)
Age
Smoker
non−smoker

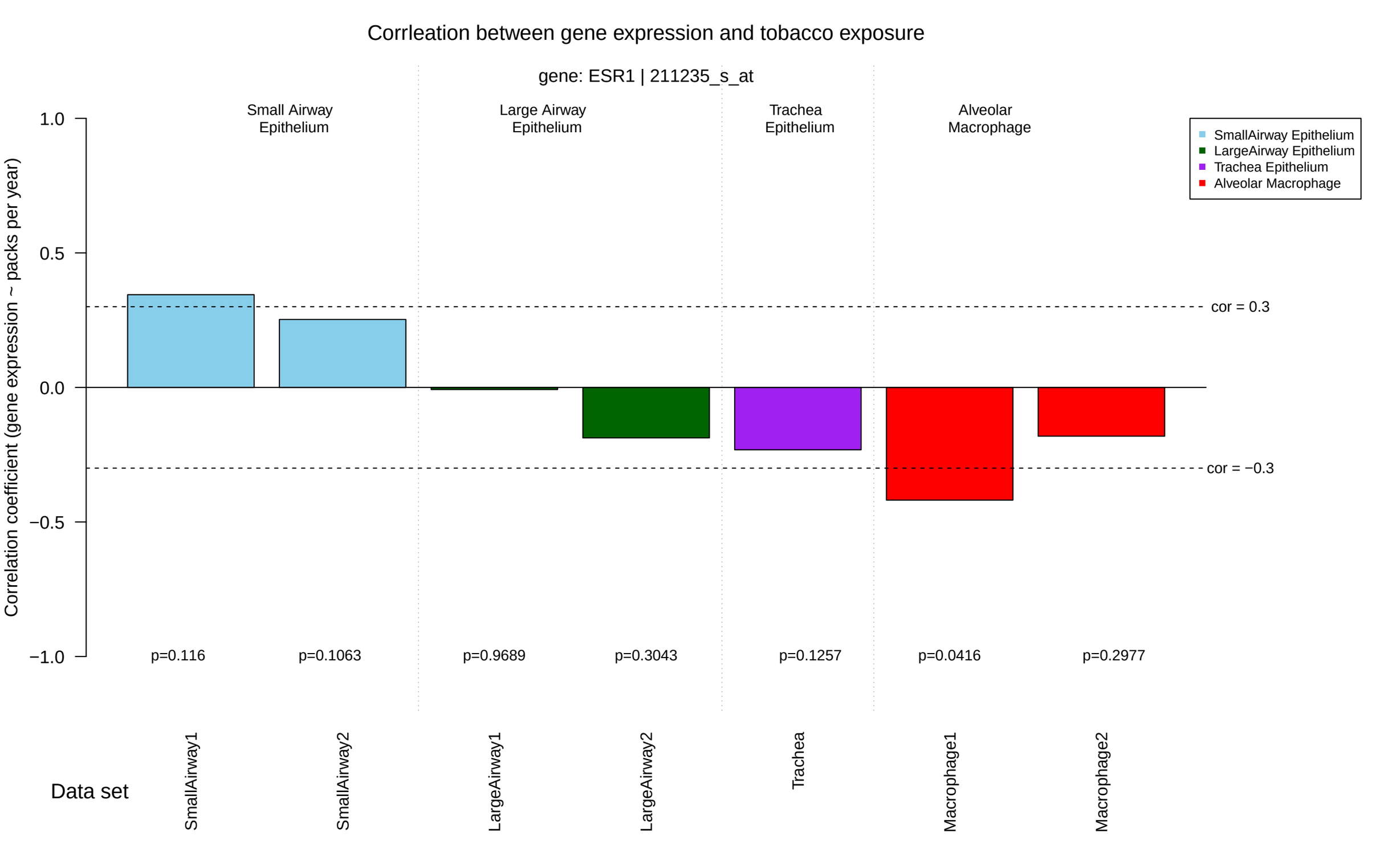

set: LargeAirway1 (p=0.5819)
ESR1 | 211235_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9597)
ESR1 | 211235_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5882)
ESR1 | 211235_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.091)
ESR1 | 211235_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8772)
ESR1 | 211235_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ESR1 | 211235_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.116
p=0.1063
p=0.9689
p=0.3043
p=0.1257
p=0.0416
p=0.2977
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2